

Generalization and Regularization

Making deep networks perform well on unseen data

Prof. Nipun Batra

ES 667 · Deep Learning · IIT Gandhinagar

The generalization problem

Where we are

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Backprop gives the gradient; the optimizer drives $\mathcal{L}_{\text{train}}$ down. We can now make the **training loss** small.

Does low training loss alone guarantee that a model will perform well on new data?

A. Yes; training loss is the only metric that matters

B. Yes, if the model has enough parameters

C. No; it may have memorized the training set

D. No; because gradient descent cannot reduce loss

Correct: C — It may have memorized the training set

Why: We care about expected performance on unseen data. A small training loss can coexist with a large generalization gap.

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We only ever **see finite data**; $\mathcal{L}_{\text{train}}$ is a noisy estimate of the quantity we actually want, $\mathcal{L}_{\text{test}}$.

The generalization gap

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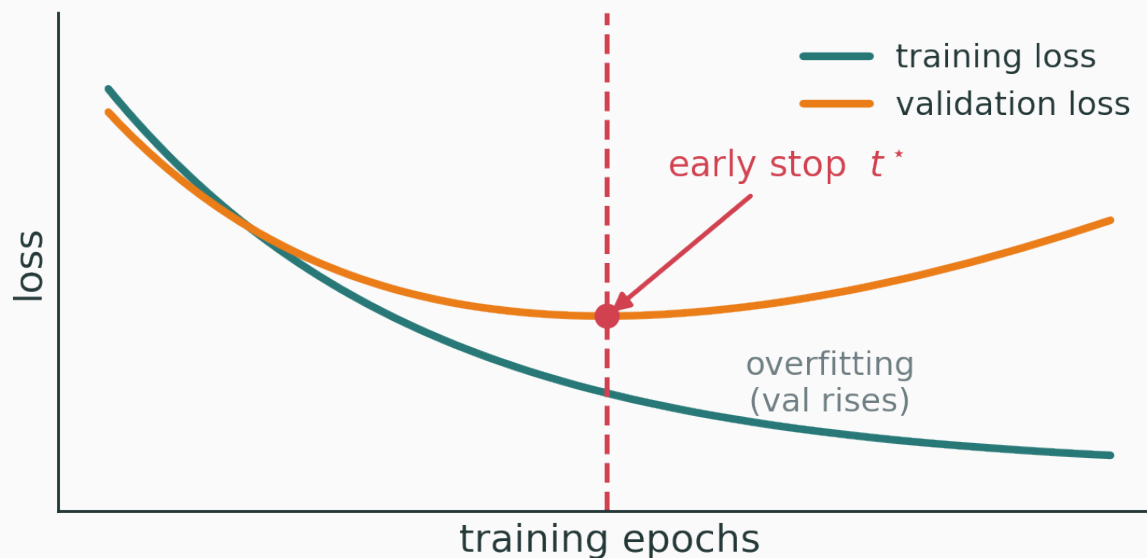
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Validation remains at chance: fitting the training set is **not** evidence of a learned, general rule.

The plan for this lecture

We have networks that train. Now we must stop them from **memorizing**.

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Lever	What it controls
splits + early stopping	when to stop
weight decay / ℓ_2	size of the weights
dropout	co-adaptation of units
data augmentation / mixup	invariance, linearity
label smoothing	overconfidence
implicit regularization	SGD noise, architecture, init

Data splits and early stopping

Three splits, three jobs

Split	Used to	Touches weights?
train	fit parameters θ	yes — via gradients
validation	select model / hyperparameters	no — only choices
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Never tune on the test set. Every peek leaks information and inflates your reported number.

Early stopping

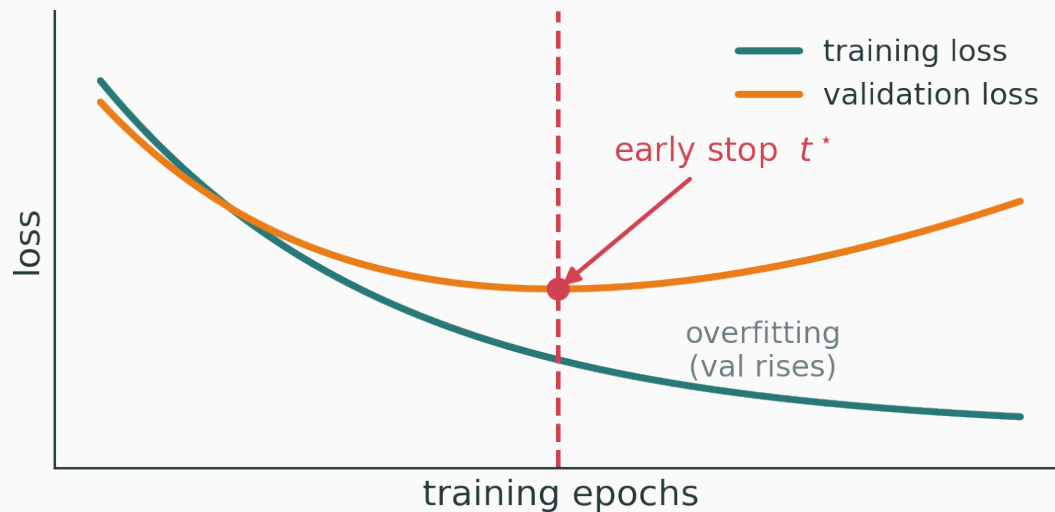
Monitor validation loss each epoch; keep the **best** checkpoint, not the last:

$$t^* = \arg \min_t \mathcal{L}_{\text{val}}(\theta_t)$$

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train keeps falling; validation is U-shaped — stop at the bottom.

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$\mathcal{L}_{\text{train}}$	1.20	0.80	0.45	0.32	0.28	0.25
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keep θ_4 — the last epoch is the wrong choice

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Early stopping limits **effective capacity** without changing the architecture — a free, always-on regularizer. It is the cheapest one you have.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/double-descent

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Find the checkpoint with the lowest **validation** loss rather than the last or lowest-training-loss checkpoint.

Capacity, bias and variance

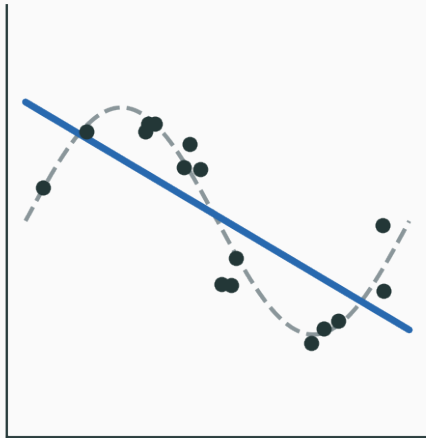
Three regimes

Regime	$\mathcal{L}_{\text{train}}$	$\mathcal{L}_{\text{val}}$	Diagnosis
underfit	high	high	too little capacity
good fit	low	low	capacity ~ right
overfit	low	high	memorizing train

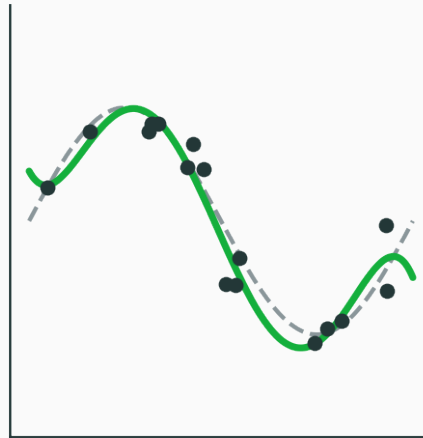
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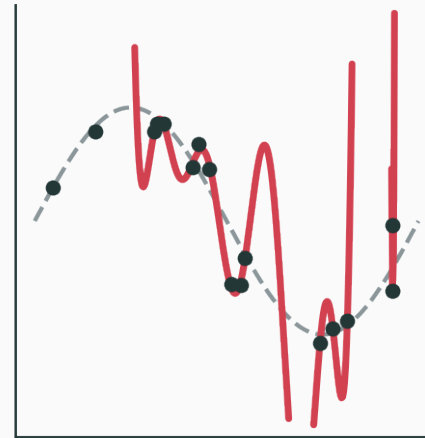
degree 1 — underfit



degree 5 — good



degree 15 — overfit



Bias–variance decomposition

Expected test error splits into three pieces:

$$\mathbb{E}[(\text{error})^2] \approx \underbrace{\text{bias}^2}_{\text{too simple}} + \underbrace{\text{variance}}_{\text{too sensitive}} + \underbrace{\sigma^2}_{\text{irreducible noise}}$$

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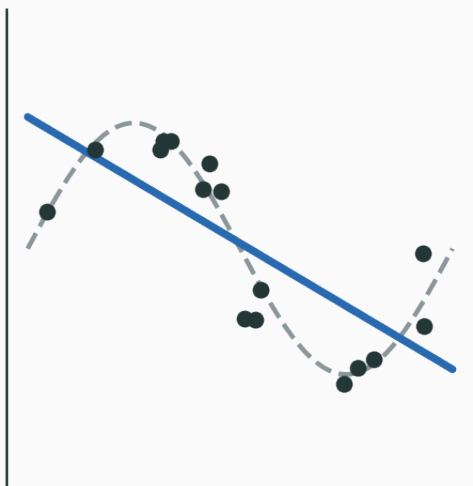
More capacity ↓ bias but ↑ variance. Classically we trade one against the other.

Fit $y = \sin(2\pi x) + \varepsilon$ from noisy samples with polynomials of degree d :

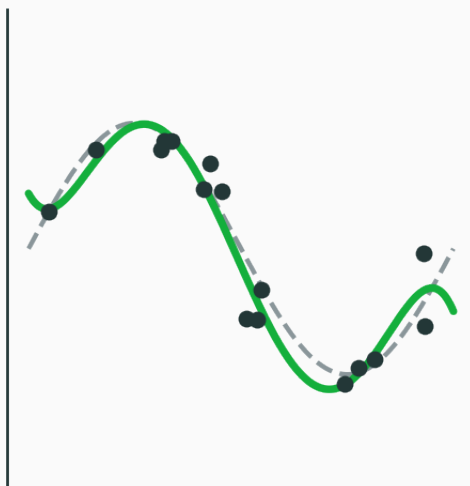
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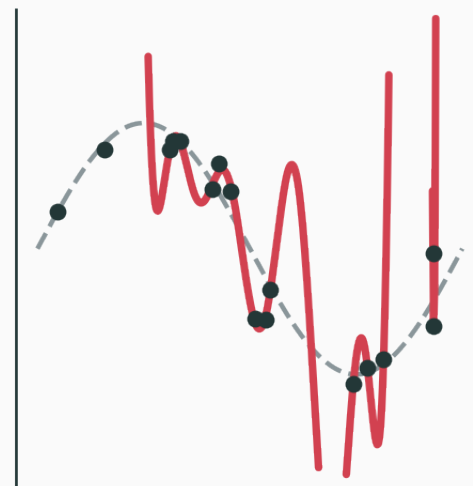
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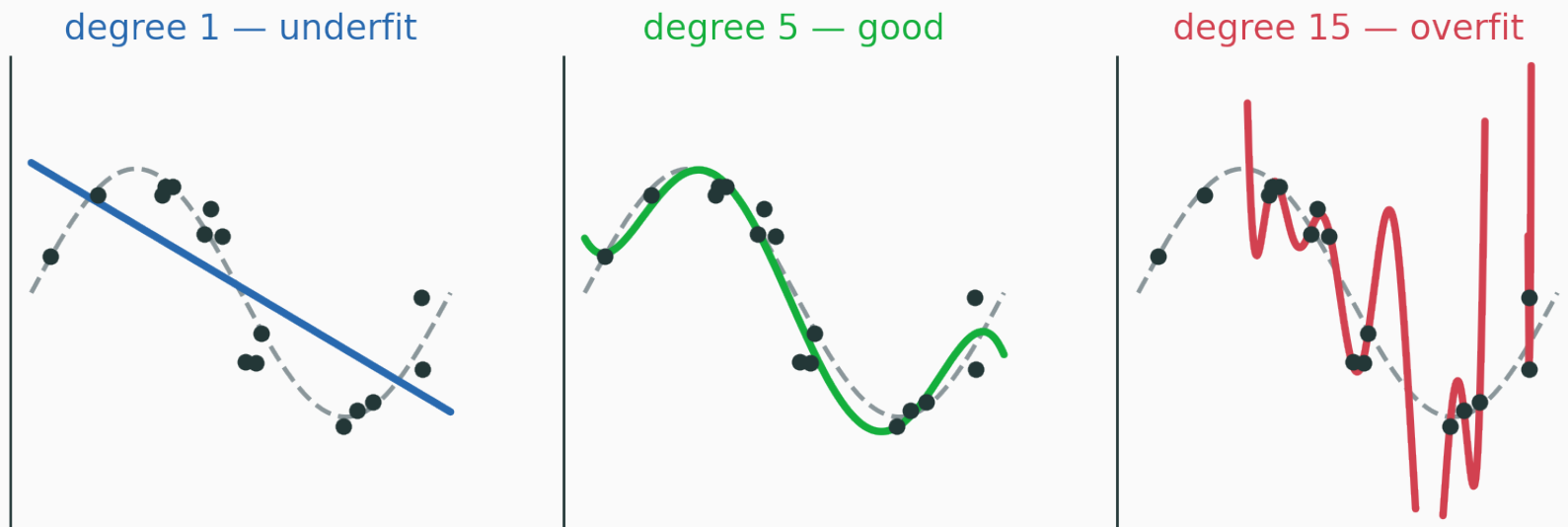


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$d = 1$ underfits (high bias) · $d = 5$ fits · $d = 15$ overfits (high variance)

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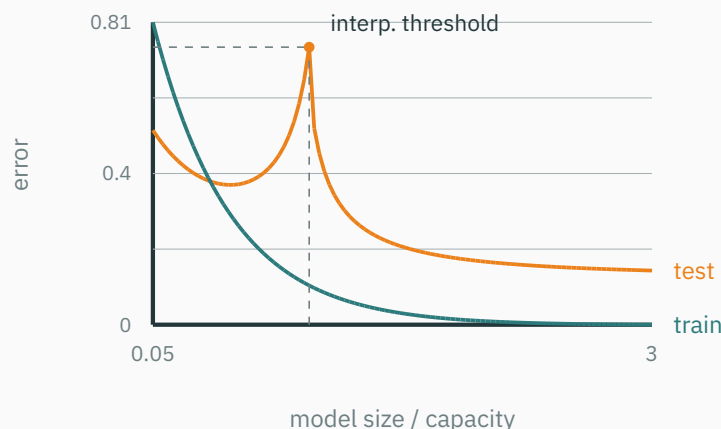
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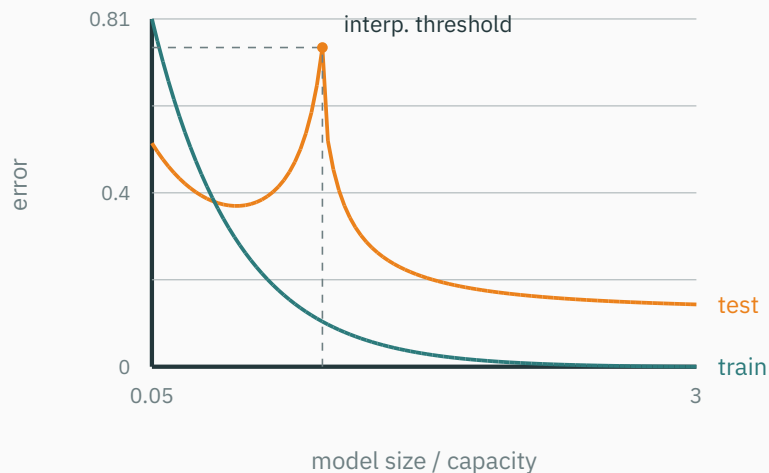
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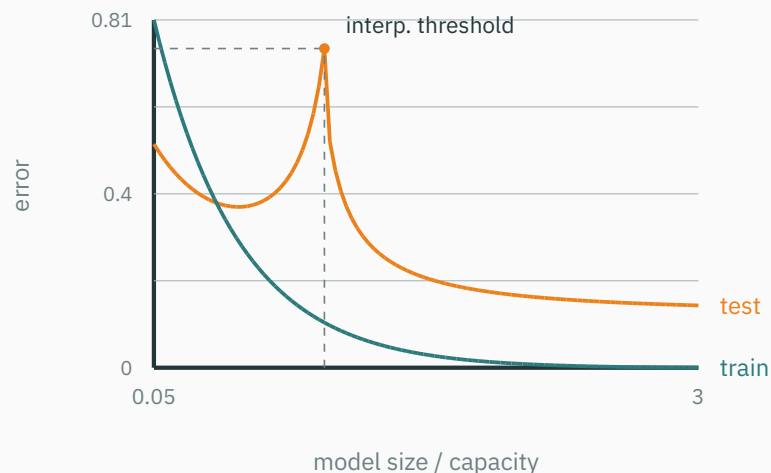
Double descent



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But **validation discipline is still non-negotiable** — it is how we know what is happening.

Weight decay and ℓ_2 regularization

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ℓ_2 regularization $\Leftrightarrow$ a **Gaussian prior** on the weights: MAP estimation with $\theta \sim \mathcal{N}(0, \sigma^2 I)$ gives exactly this penalty.

The gradient of the penalty

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a multiplicative shrinkage $(1 - 2\eta\lambda)$ **before the gradient step**

Why “weight decay”?

Each step first **shrinks** the weight toward zero, then applies the gradient (write $\lambda' = 2\lambda$ to fold the 2 into the decay rate):

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Absent any gradient, weights **decay** geometrically toward 0 — hence the name. ℓ_2 regularization and weight decay are the same mechanism (for SGD).

Worked example: one decay step

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the extra -0.1 (4.7 vs 4.6) is the shrinkage pulling the weight toward zero.

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use **AdamW**, not Adam with naive ℓ_2

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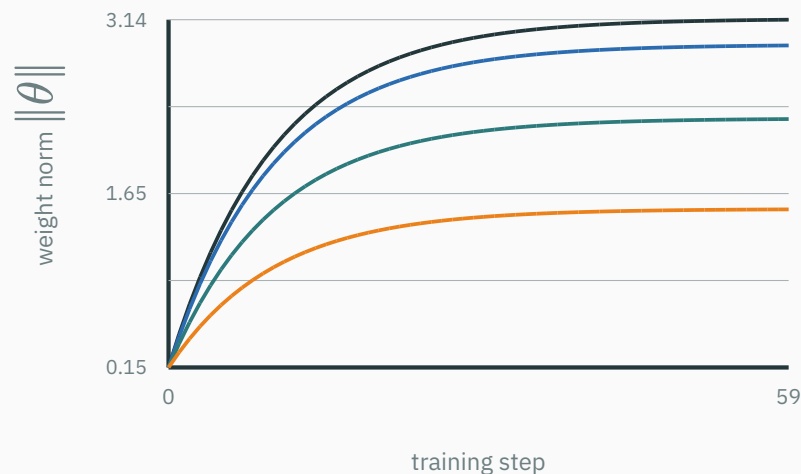
Rule of thumb: decay the **weight matrices**, leave the **one-dimensional** parameters alone.

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$$\lambda = 0 \cdot \lambda = 0.01 \cdot \lambda = 0.05 \cdot \lambda = 0.15$$

larger $\lambda \Rightarrow$ smaller final weights $\Rightarrow$ smoother function.

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Compare the unchanged shrink factor with the doubled data-gradient step when η changes.

Dropout

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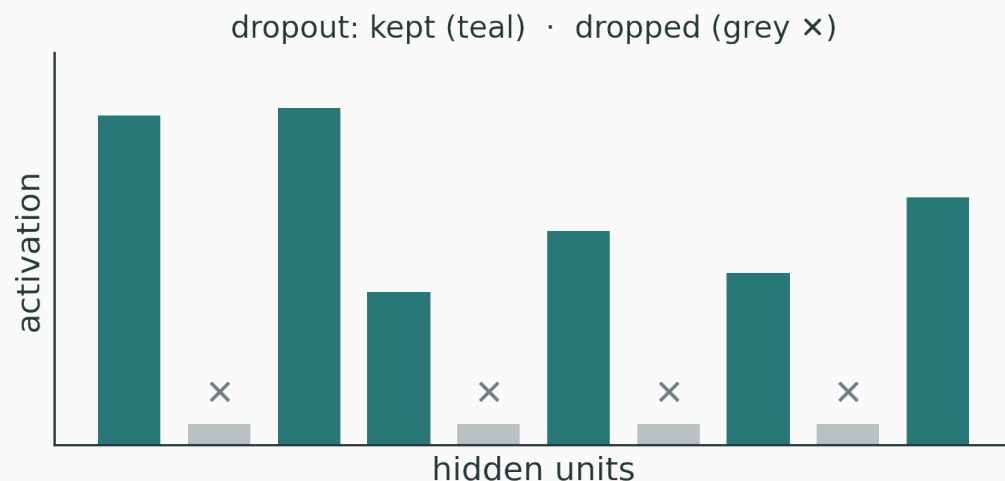
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$$\mathbb{E}[\tilde{h}_j] = \frac{1}{p} \cdot (p \cdot h_j + (1 - p) \cdot 0) = h_j$$

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$$\mathbb{E}[\tilde{h}_j] = \frac{1}{p} \cdot (p \cdot h_j + (1 - p) \cdot 0) = h_j$$

no test-time rescaling needed — at test time just use the full network

Worked example: inverted dropout

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$$\tilde{h} = \frac{m \odot h}{p} = \frac{[2, 0, 6, 0]}{0.5} = [4, 0, 12, 0]$$

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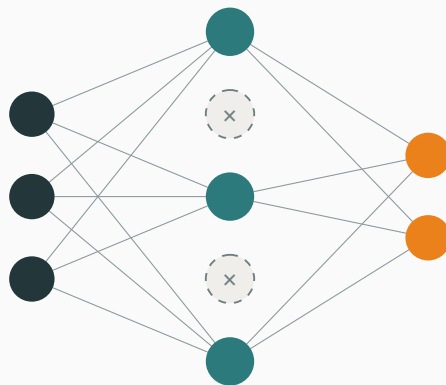
surviving units are **doubled**, so the layer's expected output is preserved.

Why it helps: model averaging

Each mask trains a **different subnetwork** — dashed units are dropped this step:

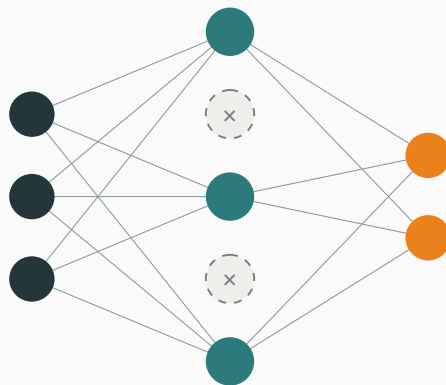
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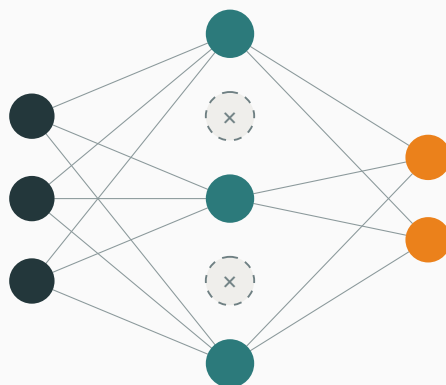
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An **ensemble-like** effect from one model — but an intuition, not an exact Bayesian average. Do not overclaim it.

Dropout caveats

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- tune the **keep** rate $p \approx 0.5\text{--}0.8$ (i.e. drop 20–50%).

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/dropout-playground

Toggle dropout on a small MLP, sweep the rate, and watch training vs validation loss — see how too much drop underfits and too little overfits.

Data augmentation

Regularize by enlarging the data

Generate new **label-preserving** examples from the ones you have:

$$(x, y) \mapsto (T(x), y), \quad T \text{ preserves the label}$$

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Domain	Transforms T
images	crop, flip, rotate, color jitter, noise
audio / time series	time-shift, mask spans, add noise
text	paraphrase, synonym swap, back-translation

If T preserves the label, we want the model to be **invariant** to it:

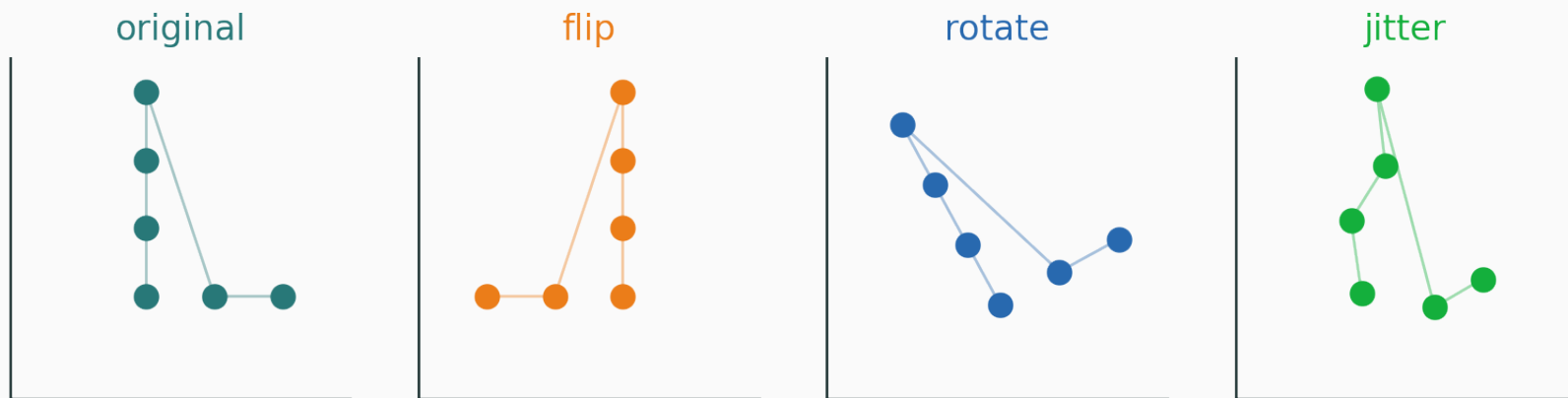
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Augmentation teaches invariance

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$$f_{\theta}(x) \approx f_{\theta}(T(x))$$

same label under label-preserving transforms



same object, many views — the network learns to ignore the nuisance transform.

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T must preserve the label. Choosing T **is** a modelling decision about what should not matter.

Mixup: interpolate examples

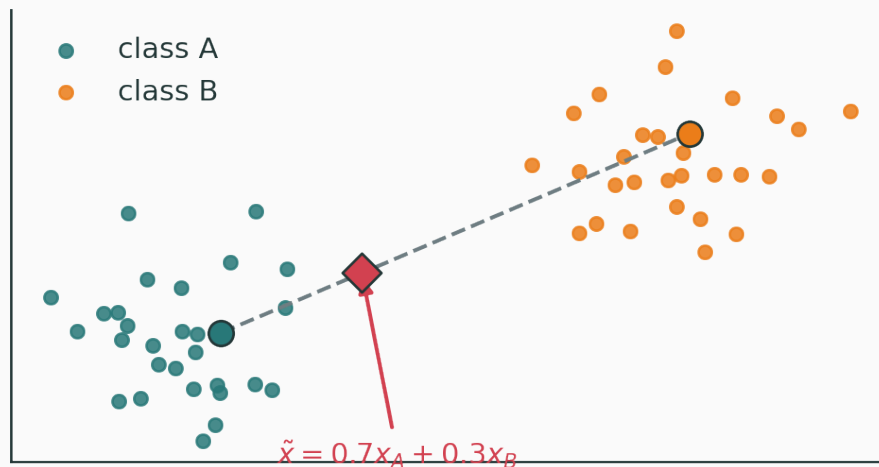
Blend two examples **and** their labels with $\lambda \sim \text{Beta}(\alpha, \alpha)$:

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j, \quad \tilde{y} = \lambda y_i + (1 - \lambda)y_j$$

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convex combinations encourage **linear** behaviour between examples.

$x_A = \text{cat}, y_A = [1, 0]; x_B = \text{dog}, y_B = [0, 1]; \text{mix } \lambda = 0.7:$

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the network is asked to be 70% cat, 30% dog — no hard boundary jump.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/augmentation-mixup

apply transforms and mixup to a 2-D dataset and watch the decision boundary become smoother and more invariant.

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Observe how soft targets discourage infinite-margin, overconfident logits.

Label smoothing

Hard labels push to extremes

With a one-hot target, cross-entropy keeps pushing $p_y \rightarrow 1$:

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The model is rewarded for **ever more extreme** confidence — even on ambiguous or mislabelled examples. This encourages **overconfidence**.

Move a little mass ε off the correct class onto the rest:

$$y_k^{\text{smooth}} = \begin{cases} 1 - \varepsilon & \text{if } k = y \\ \varepsilon / (K - 1) & \text{otherwise} \end{cases}$$

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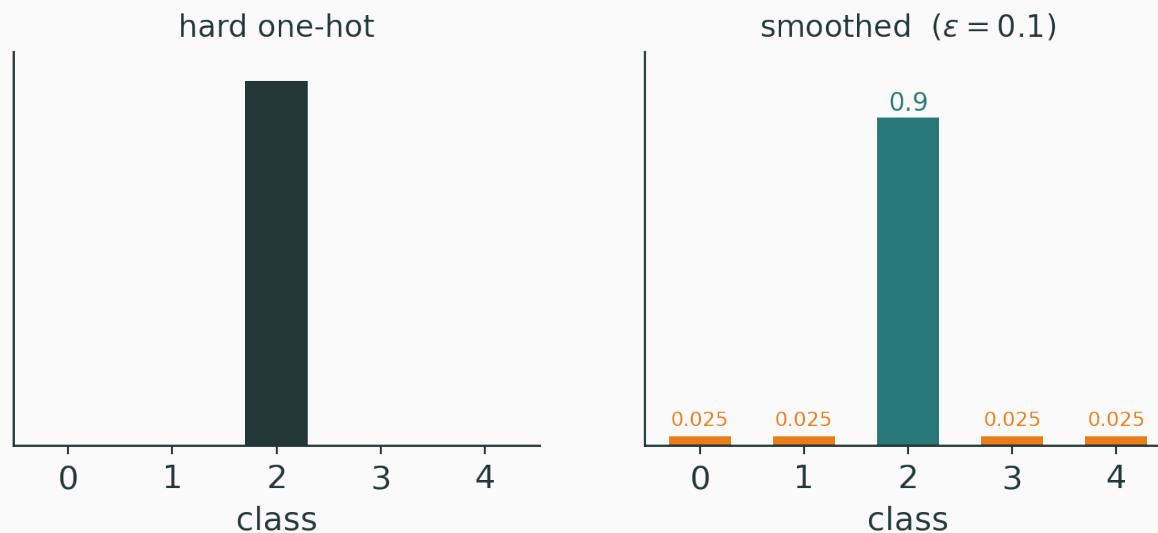
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still sums to 1; the target is achievable with **finite** logits.

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Can **hurt** when you need exact confidences — e.g. **knowledge distillation** or when downstream code reads calibrated probabilities.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/calibration

Sweep ε and watch the reliability diagram — see over-confident predictions pull back toward the diagonal as smoothing increases.

Noise and implicit regularization

Noise is a regularizer

Many effective regularizers work by **injecting noise** during training:

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Noise prevents the network from building **brittle**, precise dependencies — it must be robust to the perturbation.

Explicit vs implicit regularization

Explicit (a term / operation)	Implicit (a side effect)
$+\lambda\ \theta\ ^2$ penalty	SGD gradient noise
dropout	architecture / optimizer bias
data augmentation	early stopping
label smoothing	normalization, initialization

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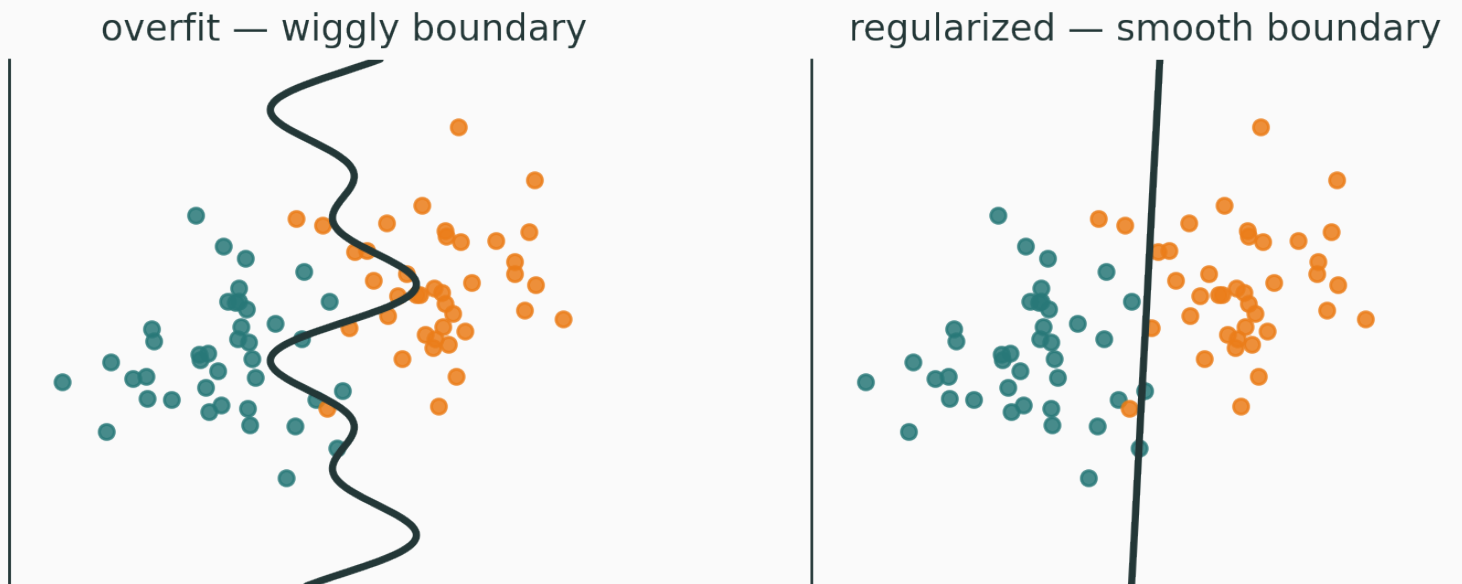
Regularization is **not only a penalty term**.

Regularization changes the learned function

Two models can both fit the training set to $\mathcal{L}_{\text{train}} \approx 0$ — yet differ:

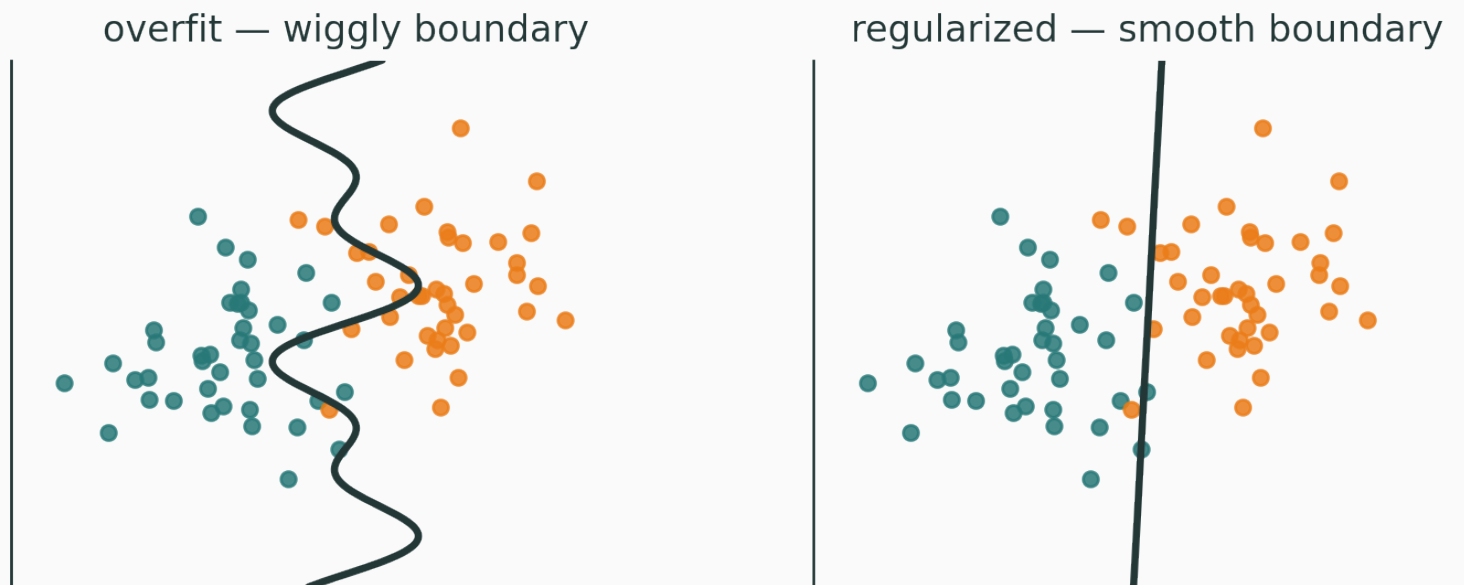
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regularization **picks** the smoother boundary — smaller norm, more invariance, less confidence, better val.

A practical workflow

Both training and validation loss are high. What should be tried before adding dropout?

A. Add stronger regularization immediately

C. Stop training earlier

B. Improve the fit: capacity, optimization, data checks, or training time

D. Increase label smoothing until training loss vanishes

Correct: B — Improve the fit first

Why: High training loss signals underfitting or a pipeline issue. More regularization would make fitting even harder.

Fix the fit first: **bigger model, train longer, better LR, less augmentation, check the labels.**

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Adding regularization to an underfit model makes it **worse**. Regularize only once you can **overfit**.

Then: regularize an overfit model

Training loss **low**, validation loss **high** $\Rightarrow$ overfitting. Now reach for:

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Lever	First choice when...
more data / augmentation	cheap to collect more or transform
weight decay	almost always — near-free
dropout	MLP / classifier head
early stopping	always on
smaller model	data is truly limited
label smoothing	over-confident / noisy labels

A regularization dashboard

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If the **gap** widens while train keeps improving, you are overfitting — turn a regularization knob.

Model	$\text{acc}_{\text{train}}$	acc_{val}	Action
A	65%	63%	underfit → add capacity , train longer
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- **A**: small gap, both low $\Rightarrow$ the model **cannot fit** — do **not** regularize;
- **B**: large gap $\Rightarrow$ the model **memorized** — rein it in.

Practical defaults

Setting	Default recipe
images (CNN)	augmentation + weight decay + early stopping
MLP / tabular	weight decay + dropout + early stopping
Transformer	AdamW + dropout + (label smoothing) + aug
small data	transfer learning + strong validation

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Start simple, add one knob at a time, and let **validation** decide whether it helped.

Summary

One table: every method

Method	What it controls
early stopping	effective complexity — when to stop
weight decay	parameter norm $\ \theta\ $
dropout	co-adaptation / activation noise
data augmentation	invariance to nuisance transforms
mixup	linearity between examples
label smoothing	overconfidence / calibration
validation split	model selection — the judge

Final mental model — one idea

Optimization **reduces training loss.**

Regularization **improves unseen-data performance.**

validation tells us whether it is actually helping.

Optimization reduces train loss; regularization improves generalization; validation is the judge.

Next: **CNNs** — why fully-connected nets are inefficient for images.